Association between isolation of Staphylococcus aureus one week after calving and milk yield, somatic cell count, clinical mastitis, and culling th...

Cows with isolation of Staphylococcus aureus approximately 1 week after calving and milk yield, somatic cell count (SCC), clinical mastitis (CM), and culling risk through the remaining lactation were assessed in 178 Norwegian dairy herds. Mixed models with repeated measures were used to compare milk yield and SCC, and survival analyses were used to estimate the hazard ratio for CM and culling. On average, cows with an isolate of Staph. aureus had a significantly higher SCC than culture-negative cows. If no post-milking teat disinfection (PMTD) was used, the mean values of SCC were 42,000, 61,000, 68,000 and 77,000 cells/ml for cows with no Staph. aureus isolate, with Staph. aureus isolated in 1 quarter, in 2 quarters and more than 2 quarters respectively. If iodine PMTD was used, SCC means were 36,000; 63,000; 70,000 and 122,000, respectively. Primiparous cows testing positive for Staph. aureus had the same milk yield curve as culture-negative cows, except for those with Staph. aureus isolated in more than 2 quarters. They produced 229 kg less during a 305-d lactation. Multiparous cows with isolation of Staph. aureus in at least 1 quarter produced 94-161 kg less milk in 2nd and >3rd parity, respectively, and those with isolation in more than 2 quarters produced 303-390 kg less than multiparous culture-negative animals during a 305-d lactation. Compared with culture-negative cows, the hazard ratio for CM and culling in cows with isolation of Staph. aureus in at least 1 quarter was 2.0 (1.6-2.4) and 1.7 (1.5-1.9), respectively. There was a decrease in the SCC and in the CM risk in culture-negative cows where iodine PMTD had been used, indicating that iodine PMTD has a preventive effect on already healthy cows. For cows testing positive for Staph. aureus in more than 2 quarters at calving, iodine PMTD had a negative effect on the CM risk and on the SCC through the remaining lactation.